

Self-Learning: ARM Processor Architecture & Features -- MPCA Module 6

Definition: The ARM (Advanced RISC Machines) processor is a 32/64-bit Reduced Instruction Set Computer (RISC) architecture engineered for high energy efficiency, low power consumption, and high performance in mobile, IoT, and embedded devices.

1. Detailed Technical Explanation

1. Key Architectural Features of ARM:

1. Load-Store Architecture: Data processing instructions (ADD, SUB, AND) operate only on registers; memory is accessed exclusively via explicit LDR (Load) and STR (Store) instructions.

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2. Register Organization of ARM (ARM7 / Cortex-A)

* Total of 37 registers (31 General-Purpose 32-bit registers and 6 Status registers), organized into banked registers across operating modes:

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3. CISC (x86/Pentium) vs RISC (ARM) Comparison

Feature	Intel x86 / Pentium (CISC)	ARM Processor (RISC)
Instruction Size	Variable (1 to 15 bytes)	**Fixed (32-bit ARM / 16-bit Thumb)**
Memory Access	Allowed directly in arithmetic ops	**Strict Load-Store Only (LDR/STR)**
Power Consumption	Higher (Desktop / Server oriented)	**Ultra Low Power (Battery / Mobile)**
Pipelining	Complex hyper-pipelines (20-31 stages)	Simple, highly efficient (3-8 stages)
Industry Dominance	PC, Laptops, Cloud Servers	**Smartphones (99% iOS/Android), IoT,

4. Quick Recall Flow

ARM RISC -> Load-Store Architecture -> Inline Barrel Shifter -> Conditional Execution -> ARM (32-bit) / Thumb (16-bit)
-> Mobile & Embedded Dominance